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The Editors
Expert Systems with Applications
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Dear Editors,

I submit for your consideration a research article entitled **"When Conformal Fusion Cannot Be Calibrated: Attainability Limits for Trace-Based Detection of Prompt-Injection Compromise."**

Large language model agents that call external tools now read email, browse the web and move money on a user's behalf. Because they ingest untrusted content and fold it into the context that drives their next action, they are exposed to indirect prompt injection: an instruction hidden in a document or a web page that the agent then follows as though the user had issued it. The applied problem this paper addresses is how an operator can watch such an agent at run time, using only the execution trace, and be given a detector that honours a false-alarm budget the operator chooses in advance.

Novel contribution relative to the published literature

Runtime monitors for such agents compute several trace-derived signals and merge them into one alarm. The merging step is universally treated as plumbing. The central claim of this paper is that it is the binding constraint, and that the constraint is a sample-size law which no published corpus in this area satisfies.

1. **An attainability theorem for dependence-robust conformal fusion.** A conformal p-value computed from n calibration values cannot fall below $1/(n+1)$. That floor caps the evidence any calibrator in the standard family $f_\lambda(p) = \lambda p^{\lambda-1}$ can express, at exactly e^{u-1}/u where $u = \log(n+1)$, attained at $\lambda = 1/u$. Averaging calibrated e-values is the only merging rule that remains valid when the signals are arbitrarily dependent—as trace signals derived from one trajectory and one judge model are—and an average never exceeds its largest term. The level-α dependence-robust test therefore has an *empty* rejection region, power exactly zero on every conceivable data set, unless $e^{u-1}/u \geq 1/\alpha$. At $\alpha = 0.05$ that first holds at $n = 312$ calibration trajectories. A companion result gives the corresponding turn-on for the Bonferroni-style rules, $n \geq K/\alpha - 1$, which is 39 for two signals. I am not aware of any prior statement of either bound, and the gap between 39 and 312 is the entire design question for a fused conformal monitor.

2. **An empirical separation of ranking from calibration.** On 90 genuinely compromised trajectories from two commercial agents, eight combination rules produce pooled AUROC within 0.010 of one another, while their a-priori true-positive rates span 0.000 to 0.867. Discrimination is nearly rule-independent; the ability to honour a stated false-alarm rate is almost entirely determined by the rule. A literature that reports AUROC alone cannot see this axis, which is why the choice has gone unexamined.

3. **The mechanism, measured rather than conjectured.** The three dependence-robust rules behave exactly as the theorems require: the e-average, mean-p and the union bound are inert, and Bonferroni is the only one that fires. Its power equals the fraction of positives carrying one signal at the p-value floor, 0.416, to three decimals—an identity, because the two signals are at the floor together in 0.000 of cases. Injection evidence in these traces is therefore dense in the bulk, where additive fusion ranks best, and sparse in the tail, where only a minimum-based rule can be calibrated. Those two facts select opposite rules.

4. **A single explanation for a standing negative result.** An earlier part of this project found that an anytime-valid e-process detector was outperformed by a hand-picked benign quantile, and reported it without an explanation. The attainability argument supplies one: time-uniform guarantees and dependence-robust merging both purchase safety with a multiplicative budget, and both need more calibration evidence than corpora of this size can express. The guarantees are sound; the corpora cannot pay for them.

5. **Supporting results on the monitor the limits were established on.** A proof that the intuitive normalisation—subtracting the largest benign value seen in training—is many-to-one and caps AUROC at one half for any attack class below that maximum, measured at a cost of 0.25 AUROC; a re-implementation of Task Shield run on the same traces with the same judge model; a benchmark-design finding, that the standard corpus delivers payloads through tool outputs a successful injection must have read and so cannot exhibit a compromise invisible to a trace monitor; and a label-polarity trap in the standard harness that inverts the compromised and resisted classes. Five components are reported as rejected on the evidence.

Fit with *Expert Systems with Applications*

The work is an applied intelligent-systems contribution. It takes a class of system now being deployed in industry, identifies a failure mode that already occurs in the wild, and delivers a monitor that an operator can actually run: it needs no model weights, no activations, no retraining and no re-execution of the agent, it costs one cached model call per distinct observation and tool call, and its false-alarm rate is calibrated rather than tuned. The evaluation is on real agent traces collected from two commercial models, not on simulation. Every number, table and figure in the paper is regenerated by released scripts from released data, without any model API access.

The journal's mandatory submission requirements

1. *Institutional e-mail addresses.* This is a single-author manuscript. My institutional address, `vinh.dq4@buv.edu.vn` (British University Vietnam), is given in the submission system and on the title page. No non-institutional address is used, so no explanation is required here.

2. *Cover letter stating the novel contribution.* Provided above.

3. *Previously rejected manuscripts, and the justification for this resubmission.* I must declare this openly. An earlier version of this work was submitted to *Expert Systems with Applications* and declined, with the comment that it fell within scope but lacked sufficient novelty. I accepted that assessment. The earlier version's positive contribution reduced, after several rounds of removing overclaims, to combining conformal signals by addition—which is Fisher's 1932 rule applied to a new domain—and the editor was right that this is not enough. The present manuscript is submitted under the journal's provision for substantially re-written work. What is new is not presentation: it is the attainability theorem and its companion turn-on result, both stated and proved here for the first time; a new study of eight combination rules that separates ranking from calibration; the measured mechanism behind it; and a new figure and table reporting all of it. The abstract, title, contributions, and conclusion are rewritten around that result, and the previous headline—a monitor with a modest AUROC gain—is now a supporting section. I would rather state the history plainly and let the editor judge whether the change is sufficient than have it discovered.

4. *Approval by all authors.* As sole author I approve this manuscript for consideration, and will confirm approval through the link sent on submission.

I confirm that the manuscript is original, has not been published previously, and is not under consideration elsewhere. There are no competing interests and no external funding was received. A blinded manuscript is supplied with all author and affiliation details removed, including the repository address, which names me; that address is given on the title page and will appear in the published version.

Thank you for your consideration.

Sincerely,

Quang-Vinh Dang